10.4 Exercises

Use the following exercises to improve your knowledge of the material:

- *Exercise 1*—Create Direct Prompting, Few-Shot Prompting, and Zero-Shot Prompting

Objective —Create three different prompts for an LLM to summarize a recent scientific article: one using direct prompting, one with few-shot prompting, and the last employing zero-shot prompting.

Tasks:

  - Compare the effectiveness of the summaries generated by each approach.
  - Compare the accuracy of the summaries generated by each approach.
- *Exercise 2*—Craft Reasoning Prompts

Objective —Design a set of prompts that require the LLM to solve logical puzzles or riddles.

Tasks:

  - Focus on how the structure of your prompt can influence the LLM's reasoning process.
  - Focus on how the same can influence the correctness of its answers.
- *Exercise 3*—Evaluation Prompt Techniques

Objective —Develop an evaluation prompt that asks the LLM to predict the outcome of a hypothetical experiment.

Task:

  - Create a follow-up prompt that evaluates the LLM's prediction for accuracy and provides feedback on its reasoning process.